

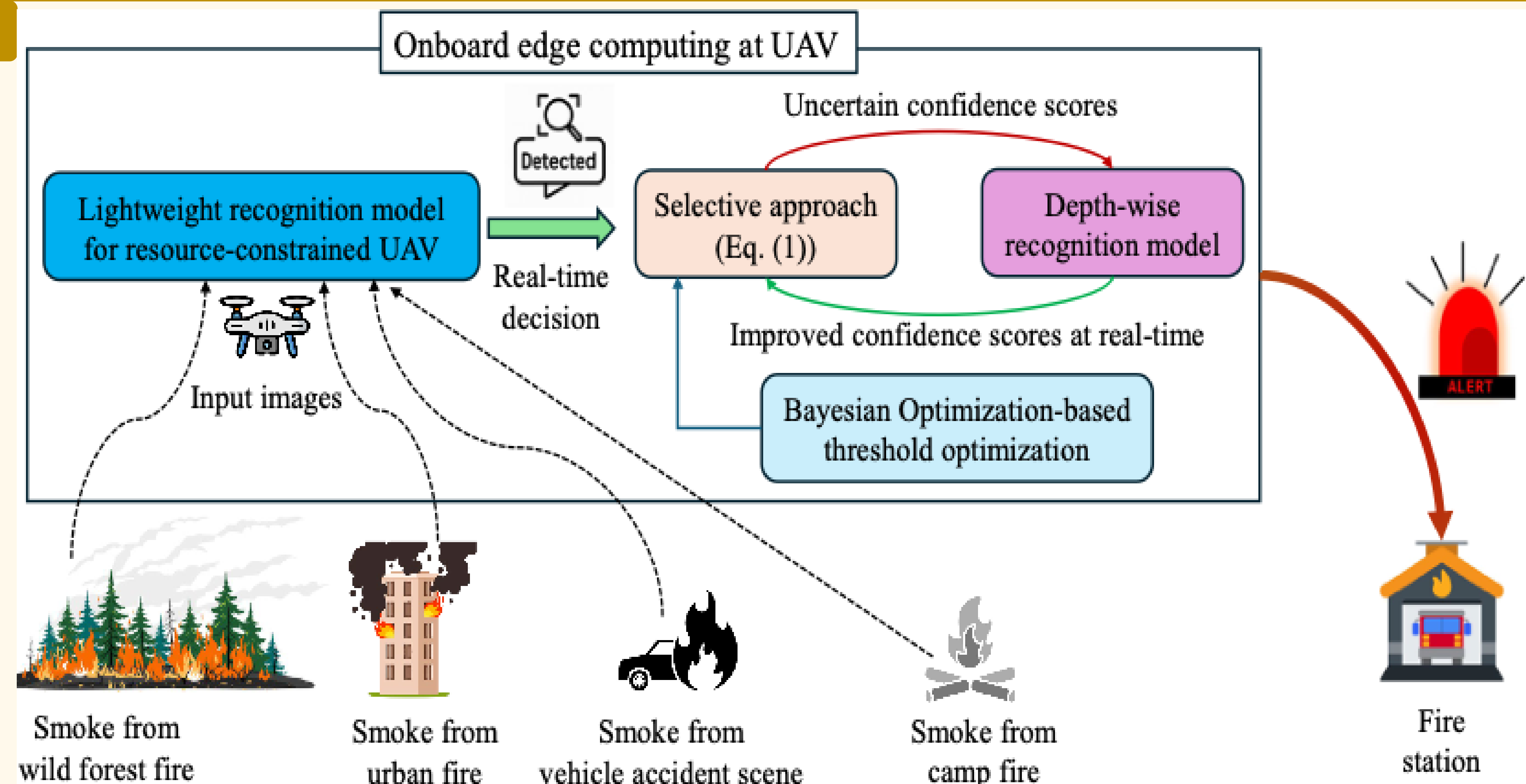
Motivation

Early wildfire smoke detection is crucial for protecting lives and reducing environmental damage. Deep learning models offer high accuracy but demand computing power and energy, limiting use on battery-powered UAVs. This work proposes a hybrid framework that maintains real-time accuracy while reducing computational costs.



Proposed Framework

The framework uses a lightweight CNN for fast image analysis, forwards uncertain cases to a stronger depth-wise CNN, and applies Bayesian Optimization to set confidence thresholds. All inference runs onboard the UAV, reducing latency and supporting remote autonomous operation.



Methodology

The framework combines:

➤ Lightweight Model

- Custom CNN with only 0.69 million parameters
- Fast inference for every incoming frame
- Provides confidence scores for smoke detection

➤ Selective Hybrid Decision

Instead of processing every image with a large network, the framework activates the depth-wise model only when the lightweight model is uncertain. The hybrid decision is defined in Eq. (1):

$$f(I, T_L, T_H | M_l, M_d) = \begin{cases} M_l(I) & M_l(I) \leq T_L \parallel M_l(I) > T_H \\ M_d(I) & \text{otherwise} \end{cases}$$

where (M_l) is the lightweight model and (M_d) is the depth-wise model.

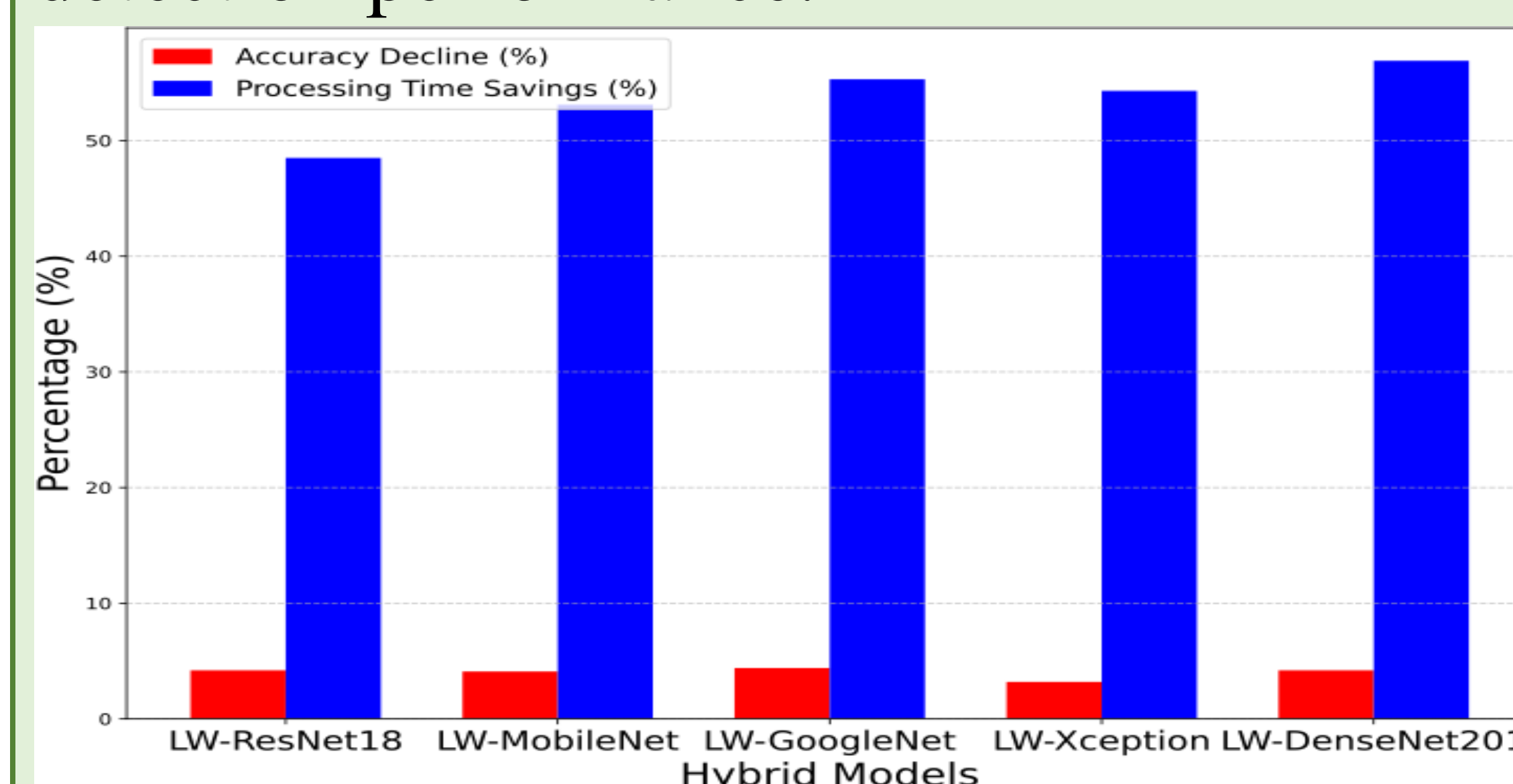
➤ Bayesian Optimization

Bayesian Optimization selects the best lower and upper thresholds to balance smoke-detection accuracy with fast processing time.

Results

Hybrid Framework Performance

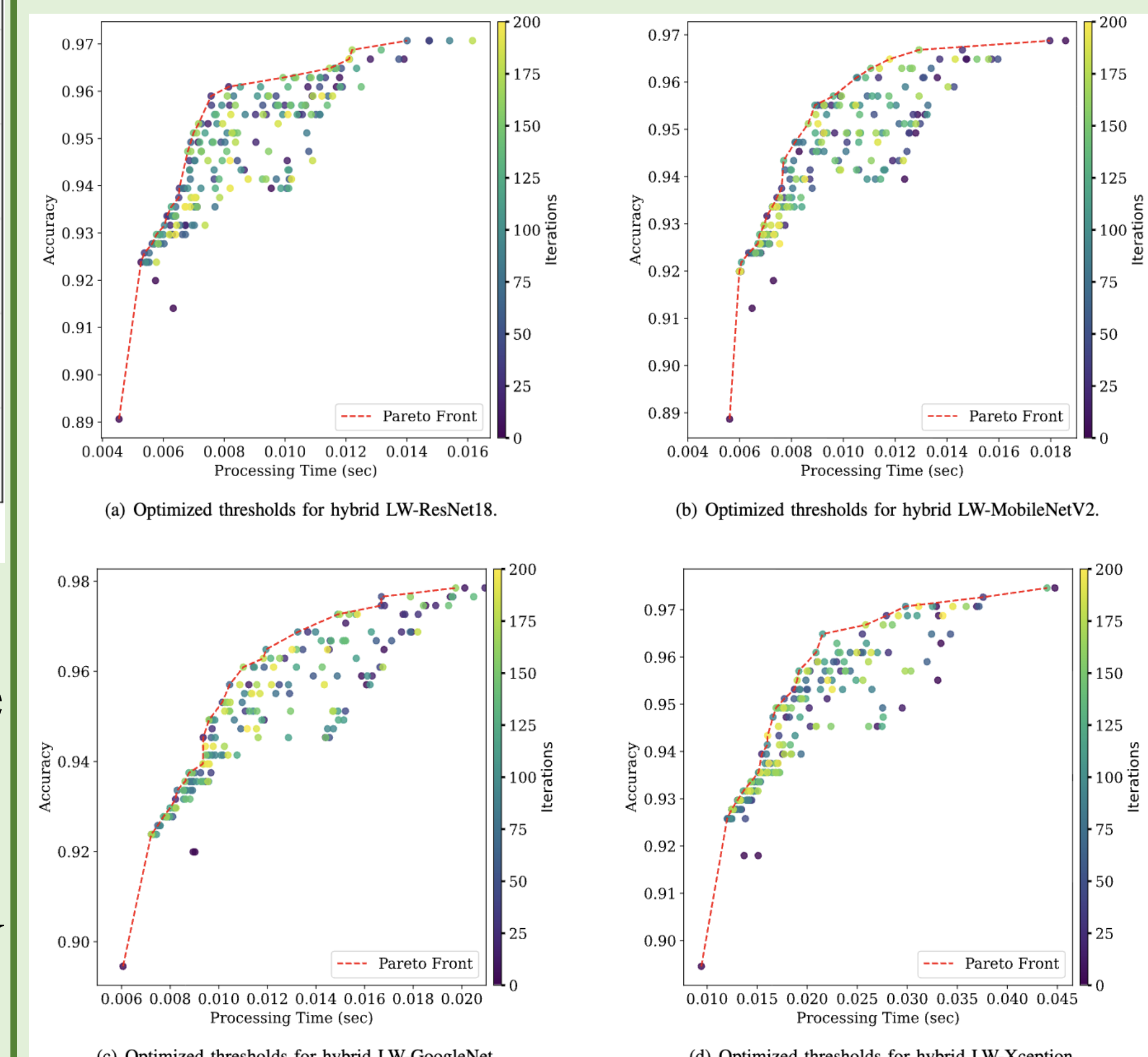
The proposed framework reduces computation while maintaining high detection performance.



- Up to **56.9%** reduction in processing time.
- Up to **64.1%** fewer calls to the depth-wise model.
- Only $\approx 4\%$ average accuracy reduction.
- Significant energy savings for longer UAV operation.

Bayesian Optimization Results

Bayesian Optimization automatically selects thresholds that balance accuracy and processing efficiency across different backbone networks.



Conclusion

- Hybrid inference enables fast and accurate onboard smoke detection.
- Bayesian Optimization automatically balances accuracy and efficiency.
- The proposed framework reduces computation and energy consumption, making it suitable for real-time UAV deployment.